

Probability in Engineering

Presented by

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Basic Probability Concepts

1

Classical Probability

$P(E) = n(E)/n(S)$; = number of favourable outcomes / number of total outcomes

2

Conditional Probability

Probability of occurrence of event A given that event B has already occurred is known as conditional probability.

$P(A|B) = P(A \text{ intersection } B) / P(B)$; provided $P(B)$ not equal to 0.

3

Random Variable

Unpredictable quantities that take on numerical values based on the outcome of an experiment.

4

Probability Distribution

Functions that describe the likelihood of different outcomes of a random variable.

5

Expected Values

The theoretical long-run average value of a random variable.

Random Variable and its Probability Distribution

Random Variable:

Let S be the sample space associated with given experiment. The real valued function 'X' whose domain is S is called a random variable.

Probability Distribution Function:

If a random variable takes value $x_1, x_2, \dots, x_n$ with respective probabilities $P_1, P_2, \dots, P_n$. Then

$X : \quad \quad x_1 \ x_2 \ \dots \ x_n$

$P(X=x_i) : P_1 \ P_2 \ \dots P_n$

is called Probability Distribution Function of 'x'.

Example:

Two coins are tossed simultaneously. Let 'X' be the number of tails appeared then $S = \{HH, HT, TH, TT\}$ $X =$ No. of tails appeared. Here X is a random variable & its PDF is

$X : \quad \quad 0 \ 1 \ 2$

$P(X=x_i) : \frac{1}{4} \ \frac{2}{4} \ \frac{1}{4}$

Bernoulli Trials and Binomial Distribution

Bernoulli trial

Trials of a random experiment are called Bernoulli trials if

- (a) There are finite number of trials (b) Trials are independent (c) Each trial has exactly two outcomes: Success or failure (d) Probability of success remains same in each trial

Binomial Distribution

It can be easily observed that when Bernoulli's trial is repeated 'n' times, then probability of getting 'r' successes and n – r failures is given by

$$P(X=r) = {}^nC_r (p)^r (q)^{n-r}$$

Where random variable 'X' is representing number of successes, p is probability of success & q is probability of failure Clearly $p + q = 1$

Example

A box contains 24 identical balls of which 12 are white and 12 are black. The balls are drawn at random from the box one at a time with replacement. Find the probability that a white ball is drawn for the fourth time on 7th draw?

Here, we need 7th drawn ball to be white, such that before it 3 white balls have been drawn out of previous 6 draws Using binomial distribution Probability of drawing white ball $p = 12/24 = \frac{1}{2}$ = Probability of drawing black ball q.

Required probability = $({}^6C_3 p^3 q^3) \frac{1}{2} = \frac{5}{32}$

Applications of Probability in Engineering

1

Reliability Analysis
Assessing the ability of a system or component to function under specific conditions and probabilities of failure.

2

Quality Control
Controlling processes and minimizing the probability of defects in production systems.

3

Decision Making
Assisting in making informed decisions based on the estimated probability of different outcomes.

Real-World Case Studies



Tacoma Narrows Bridge Collapse

An example of how ignoring probability and failure analysis can lead to catastrophic results.



Aircraft Design

How probability theories help reduce the likelihood of accidents and improve safety in the aviation industry.



Financial Risk Management

How banks and financial institutions use probability models to predict and mitigate the risk of financial losses.

6. Conclusion

Probability provides engineers with a powerful framework to tackle uncertainties and make informed decisions. By understanding and applying probability theory, engineers can enhance system reliability, optimize designs, and minimize risks. Embracing a statistical perspective in engineering enables better problem-solving and ensures the success of complex projects.

THANK YOU!